

LRN Analytics — Denial & AR Workbench

Developer Handoff: AR Queues, Claim-Level Logic & Workflow Enhancements

Session decisions (28 September 2026) + detailed functional requirements from the Workbench mockup and Build Reference

Item	Detail
Document version	v1.1 — Draft for review (adds detailed requirements from mockup & Build Reference review)
Prepared	29 September 2026 (v1.0); updated 29 September 2026 (v1.1)
Sources	AR WorkBech Plan and Clarification.docx (meeting transcript, 1h 28m); LRN_Denial_AR_Workbench_Demo_Account.html (functional mockup — source code reviewed); AR_Workbench_Build_Reference.pdf
Session participants	Banuka Jayakodi (business owner / RCM), Azwin Sarap (RCM operations), Prasanna R (platform build)
Related artifacts	ARWorkflow_Redesign_Plan_v1.0_09.25.2026.xlsx (build plan, packaged with this document); ARWorkBench_Clarifications_v1.0.docx; ARWorkbench-Statuses-Queues-v1.0_09.25.2026.xlsx
Target completion	13 November 2026 (agreed in session)
Status	Pending review by RSKP before release to development

How to read this document: Items marked **Confirmed** were agreed in the session. Items marked **To confirm** were discussed but not fully closed, or contain an interpretation of the transcript that should be validated before build. All open questions are consolidated in Section 13.

Contents

Document Structure & Revision History	5
1. Executive Summary.....	6
2. Data Sync & Data Grain.....	6
2.1 Master file sync.....	6
2.2 Denial insights linkage	6
2.3 Claim detail view.....	7
3. AR Queue Logic.....	7
3.1 Queue definitions	7
3.2 Rules and configuration	8
4. Claim-Level Denial Logic	8
4.1 Primary denial code	9
4.2 Code format standard.....	9
4.3 Denial categorization	9
5. Weekly Re-Sync Behaviour	9
6. Auto Adjustments	10
7. Claim Financial Fields.....	10
8. Follow-Up Workflow.....	10
8.1 Follow-up notes and dates.....	10
8.2 Attachments and Document Vault	11
8.3 Write-off approval	11
8.4 Internal escalation and reassignment requests.....	11
8.5 Batch assignment due date.....	11
9. Client Escalation (CIP) Workflow	11
9.1 Flow.....	12
9.2 Comment visibility	12
10. Roles, Access & User Management	12
10.1 Client-level user provisioning.....	13
11. New Features & Enhancements Summary	13
12. Delivery Plan & Priorities	13
13. Open Items & Decisions Pending.....	14
14. Suggested Acceptance Tests.....	14
15. Roles, Permissions & Data Scoping (FR-ACC).....	16
15.1 Roles.....	16
15.2 Permission matrix (server-enforced)	16
15.3 Requirements.....	16

16. Navigation Map & Notification Badges (FR-NAV)	17
17. Data Processing & Denial Analysis Insights (FR-DP)	18
18. AR Queue Classification Engine — Final Specification (FR-ARQ)	19
18.1 Lifecycle predicates	19
18.2 Workflow statuses	19
18.3 Precedence (evaluate top to bottom, first match wins)	19
19. Dashboard (FR-DASH)	20
19.1 KPI tiles	20
20. Denial & AR Work Queue (FR-WQ)	21
21. Claim Detail Workspace (FR-CD)	22
22. Log Follow-Up Note — Comments Framework (FR-FU)	23
23. Assignment Management (FR-ASG)	24
24. My Work (FR-MW)	24
25. Follow-Up Management (FR-FUM)	24
26. QA Verification Queue (FR-QA)	25
27. Escalation & Reassignment Requests (FR-REQ)	25
28. CIP Escalations — Internal Queue (FR-CIP)	26
28.1 CIP case lifecycle	26
29. Client Portal — Escalation Requests (FR-CLI)	27
30. Automatic Adjustments & Write-Offs (FR-ADJ)	27
31. Recovery & Financial Analytics (FR-AN)	27
32. Reports (FR-RPT)	28
33. Client Management (FR-CM)	28
34. User Management (FR-USR)	28
35. Audit Logs (FR-AUD)	29
36. System Settings — Master File Maintenance (FR-SET)	29
37. Shared Table & UI Utilities (FR-UI)	30
38. Excel Export & Bulk Status Upload (FR-XL)	30
39. Attachments & Document Vault (FR-DOC)	31
40. Weekly Re-Sync Rules (FR-SYNC)	31
41. Reconciliation Table — Do Not Copy the Mockup As-Is	32
42. Entities & Key Fields	34
42.1 Claim (claim level)	34
42.2 Child and supporting entities	34
43. Non-Functional Requirements (NFR)	36
44. Requirement → Build Plan Traceability	37
45. Requirements Not Yet Covered by a Plan Task	37

46. Additional Open Items from the Review	38
---	----

Document Structure & Revision History

Part	Contents	Sections
A	Business decisions from the 28 Sep 2026 working session (Banuka, Azwin, Prasanna)	1 – 14
B	Detailed functional requirements by module (FR-xx), derived from the mockup source code and Build Reference, reconciled with Part A	15 – 40
C	Mockup vs. session-decision reconciliation — where the mockup must NOT be copied as-is	41
D	Data model — entities, claim fields and master data	42
E	Non-functional requirements	43
F	Build-plan traceability, plan gaps and consolidated open items	44 – 46

Version	Date	Change
v1.0	29 Sep 2026	Initial handoff from the session transcript (Part A).
v1.1	29 Sep 2026	Added Parts B–F after a full review of the mockup (all 14 application scripts and seed/meta data) and the AR Workbench Build Reference.

Precedence rule for developers: Where the mockup and a Part A session decision differ, **Part A wins**. Every known difference is listed in Section 41. The mockup remains the reference for layout, fields, filters and interaction patterns not overridden there.

PART A — SESSION DECISIONS (28 SEP 2026)

1. Executive Summary

The session clarified the business rules that drive the Denial & AR Workbench. The most significant outcome is that the Workbench is **not** limited to claims tied to weekly denial insights: the **entire claim-level master file is synced** and every claim is classified into an AR Queue by rule. Denial insights remain a navigation aid that routes managers into filtered claim sets.

Key build impacts:

- **Queue engine** — rule-based classification of every synced claim into Closed, Active (auto-routed for assignment), Patient, Workflow and Special queues; driven by a configurable Non-Collectible denial list in System Settings.
- **Claim grain** — queues, listings and counts are always at claim level; line-level (CPT) detail, including line-level denials, is shown inside the claim via an expandable CPT breakdown.
- **Primary denial** — use the reporting team’s curated “Denial Code” column for categorization until a denial-hierarchy mapping is delivered; CARC group prefixes (CO/PR/PI/OA) are stripped for mapping.
- **Financial fields** — “Expected Payment” renamed to **Revenue Expectation**; Potential Recovery and Recovery Status formulas defined.
- **Re-sync behaviour** — rules for claims whose denial changes between weekly syncs, including a new **Re-Follow-Up Required – New Denials** sub-queue.
- **New capabilities** — Auto Adjustments queue with PMS export, write-off QA approval loop, attachments with a Document Vault on Azure Blob Storage, categorized escalation/reassignment reasons, and clinic-scoped client users.

2. Data Sync & Data Grain

2.1 Master file sync

- **Confirmed:** The full claim-level master file for each lab is synced (weekly cadence, aligned with the master file refresh). Classification into queues happens after sync, based on rules in Section 3.
- This supersedes the earlier assumption in ARWorkBench_Clarifications_v1.0 that only claims linked to current/previous-week denial insights are synced — which also resolves the question of how No Response claims enter the Workbench.
- Both the line-level and claim-level files are used. Dashboard metrics may draw on both; every list, queue and count **represents a claim, never a line**.
- Recovered payments are taken from the master file sync (insurance payment data is already present in the master).

2.2 Denial insights linkage

- The dashboard’s **Latest Denial Analysis Report Insights** shows the current week’s insights from the denial summary report.

- The **Data Processing** tab shows current-week plus previous-week insights, limited to items that are still unresolved.
- Clicking an insight routes the user to the filtered claim-level set in the Denial & AR Work Queue, from which the AR Manager/Team Lead can bulk-assign.
- Counts and balances on an insight cover **all payers** for that denial code. “Highest impact payer” is an informational highlight only, not a filter.
- **Confirmed:** Once claims are assigned, they must be removed from the insight’s outstanding count/balance — the insight shows only the unassigned remainder.

2.3 Claim detail view

- **Claim listing tab:** grouped at claim level.
- **Denial Info section:** shows the claim’s primary denial (code, description, category, classification).
- **CPT Line-Level Detail section:** add an **expand control per CPT** to reveal line-level information, including line-level denial codes (with original group prefix retained for display), descriptions and classifications. This mirrors the current Denial Workflow line-level view.
- Line-level data is for follow-up support only; it does not drive queue placement or categorization in this phase.

3. AR Queue Logic

Every synced claim is placed in exactly one financial-status queue. Workflow queues (QA, CIP, Re-Follow-Up) track work state on top of that. Definitions below use: **Ins Bal** = outstanding insurance balance, **Pt Bal** = patient balance, **NC List** = the Non-Collectible denial code list maintained in System Settings.

3.1 Queue definitions

Queue	Group	Entry rule	Auto-routed to work queue?
Closed – Fully Paid	Closed	Payment received; Ins Bal = 0 and Pt Bal = 0 (no balance of any kind). Includes claims meeting the “financially recovered” criteria.	No
Closed – Fully Adjusted	Closed	Claim fully adjusted; no insurance or patient balance.	No
Non-Responder	Active AR	Ins Bal > 0 and the full balance sits in the insurance bucket; no payment, no denial, no adjustment. Treated as possible collectible.	Yes
Denied – Possible Collectible	Active AR	Ins Bal > 0 and a denial received; primary denial code NOT in NC List.	Yes
Denied – Non-Collectible	Active AR (restricted)	Ins Bal > 0 and primary denial code IS in NC List (e.g., prior authorization, out of network).	No
Partially Paid – Possible Collectible	Active AR	Payment received on some CPTs and Ins Bal > 0 outstanding; primary denial NOT in NC List.	Yes
Partially Paid – Non-Collectible	Active AR (restricted)	Payment received and Ins Bal > 0; primary denial IS in NC List.	No

Queue	Group	Entry rule	Auto-routed to work queue?
Partially Adjusted	Active AR	No denial; some CPTs adjusted; Ins Bal > 0 outstanding.	Yes
Patient AR	Patient	Pt Bal > 0 and Ins Bal = 0. Denials are not evaluated for this queue.	No
Client Escalation (CIP) Pending	Workflow	Claim escalated to the client and still open.	—
CIP Response Received	Workflow	Client has responded to a CIP escalation.	—
Re-Follow-Up Required – Unresolved Claims	Workflow	Claim touched/followed up and no reply or action for 45 days since last touch.	Yes (returns for reassignment)
Re-Follow-Up Required – New Denials (NEW)	Workflow	Previously worked claim receives a new denial on a subsequent sync (see Section 5).	Yes (returns for reassignment)
Submitted for QA	Workflow	Agent has worked the claim; awaiting QA approval.	—
QA Rejected	Workflow	QA rejected; auto-assigned back to the same agent for rework.	—
Completed Claims	Workflow	QA approved.	—
Auto Adjustments (NEW)	Workflow	Claims processed by the auto-adjustment run (see Section 6).	—

3.2 Rules and configuration

- **Non-Collectible list** — maintained by the AR Manager/Admin in System Settings as a list of denial codes mapped to “Non-Collectible”. Everything not on the list is Possible Collectible. The same list drives both the Denied and Partially Paid splits.
- **Active work queue feed** — only claims with an insurance balance that are unresolved flow automatically into the Denial & AR Work Queue for assignment: Non-Responder, Denied – Possible Collectible, Partially Paid – Possible Collectible, Partially Adjusted.
- **Restricted (overall AR) queues** — the overall AR queue view, including Closed and Non-Collectible queues, is visible only to **System Administrator, AR/RCM Manager and Team Lead**. They can pull claims from any of these queues and assign them to agents for special projects (example given: a fully-adjusted claims clean-up project).
- **QA Rejected** claims are auto-assigned to the agent who worked them.

To confirm: The transcript describes the Denied and Partially Paid queues as “hidden” (restricted). The interpretation above is that the restriction applies to the overall AR queue view and the Non-Collectible/Closed queues, while Possible Collectible claims auto-feed the work queue. Please confirm which queues are restricted. The full evaluation precedence (workflow queues first, then financial state) is specified in Section 18.3.

4. Claim-Level Denial Logic

4.1 Primary denial code

- **Confirmed (interim):** Use the reporting team’s curated **Denial Code** column in the claim-level master file as the primary denial for categorization and queue placement. This column reflects the latest denial on the claim, which is what the team reports to clients.
- The claim-level file may also hold comma-separated lists of all line-level denials; these are retained for line-level display only.
- **Planned:** Banuka will deliver a denial-hierarchy mapping so the system can derive the primary/latest denial itself. Principle: terminal denials override others — e.g., if 204 (service not covered by benefit plan) or 197 (precertification/authorization absent) is present alongside another code, the terminal code is primary. Build the derivation as a configurable ranking table so the mapping can be loaded without code change.
- Mixed claims (one CPT collectible, another non-collectible) are out of scope for this phase; classification follows the claim-level primary denial.

4.2 Code format standard

- **CARC codes:** strip the group-code prefix (CO, PR, PI, OA) for all mapping, categorization and master-file lookups — e.g., “PR 204” and “CO 204” both map as “204”.
- **RARC codes:** do **not** strip — the letter is part of the code (e.g., N57, M127, MA130).
- Line-level display inside the claim may keep the original prefixed value for reference, but it must not be used for any mapping.

4.3 Denial categorization

- Banuka will provide the denial categorization for all denial codes; load it into the Denial-Action master used for classification, action and task assignment.

5. Weekly Re-Sync Behaviour

Scenario on next sync	System behaviour
Claim assigned to an agent, not yet worked , denial code changes	Update the denial on the claim silently; no reassignment. If the new primary denial is on the NC List, auto-process the claim and remove it from the agent’s queue.
Claim already worked (Submitted for QA / Completed / QA Rejected) and a new denial is received	Let the current QA step complete, then move the claim to Re-Follow-Up Required – New Denials as unassigned, so the AR Manager can reassign it (to the same or a different agent). Preserve full history so the manager and QA can see the prior touch and the new denial.
Claim followed up, no reply/action for 45 days since last touch	Move to Re-Follow-Up Required – Unresolved Claims .
Claim previously auto-adjusted or approved for write-off in the Workbench but still open in the master (not posted in the PMS)	Flag the claim, return it to the Auto Adjustments queue (or the write-off pending-posting queue) and notify the assigned agent and AR Manager.
Claim now meets closed criteria	Move to the relevant Closed queue; recovered amount recognized per Section 7.

6. Auto Adjustments

- Certain denials require adjustment rather than follow-up (examples given: 242, 197). Maintain these as a configurable **Auto-Adjust** code list.
- A **Process Auto Adjustment** action (triggered from Data Processing / the work queue) will: add an automatic system comment to each matching claim, nullify the balance within the Workbench, and move the claims to the new **Auto Adjustments** queue (instead of Completed Claims as in the current demo).
- Because the Workbench is not integrated with the PMS (AdvancedMD), the AR Manager/Team Lead must be able to **export the Auto Adjustments queue** as a report and post the adjustments in bulk in the PMS.
- If an auto-adjusted claim reappears on a later sync (adjustment not posted), apply the re-sync rule in Section 5.

7. Claim Financial Fields

Field	Definition / formula	Status
Revenue Expectation (renamed from “Expected Payment”)	Allowable amount calculated at the Medicare rate across all billed CPTs on the claim. If no rate is available, set to 0.	Confirmed rename; rate source to confirm
Potential Recovery	Revenue Expectation – (Insurance Payments + Patient Payments + Patient Balance). Floor at 0.	Confirmed
Recovery Status	In Progress — insurance payment, patient payment and patient balance are all 0. Partially Recovered — any of the three > 0 but the total < Revenue Expectation. Fully Recovered — total ≥ Revenue Expectation. No Recovery — raised in session; definition to confirm (e.g., closed/adjusted with nothing collected).	Confirmed (No Recovery to confirm)
Recovered Amount (dashboard)	Sum of insurance payments whose payment date is after the denial date, sourced from the master file sync. Payments only — adjustments excluded.	Confirmed
Financially recovered → Closed	Claims meeting the “financially recovered” criteria in the clarification document are treated as Closed.	Confirmed

8. Follow-Up Workflow

8.1 Follow-up notes and dates

- Follow-up notes are entered at **claim level only**; no CPT-level notes.
- **Next Follow-Up Date** defaults to **today + 45 calendar days** (weekends included), in line with the industry rollover standard of 45 days from last touch.

- Action-specific follow-up timelines (e.g., resubmission 30 days, appeal 60 days) and an agent-selectable “days” dropdown were discussed and **deferred**: launch with the 45-day rule, measure agent behaviour, then introduce a back-end action-to-timeline mapping in a later phase.

To confirm: Whether agents may override the default date to an earlier date (within 45 days), and whether a later date is blocked.

8.2 Attachments and Document Vault

- Enable attachments on follow-up notes for **all actions** (e.g., appeal letters submitted, EOBs received via call/fax, medical records), not only CIP escalations.
- File size limit per upload: **10–15 MB** (5 MB considered too restrictive); final value to confirm.
- Each upload requires a **document category** (proposed: Appeal, EOB, Medical Record, Payer Correspondence, CIP Response, Other).
- Build a **Document Vault** where users can search documents by claim/visit, patient and category.
- **Storage:** Azure Blob Storage (not the SQL database), with HIPAA security controls enabled — encryption at rest, private access, role-based access, and access audit logging.

8.3 Write-off approval

A manual write-off selected by an agent must pass QA before it is actioned:

1. Agent selects **Write-Off** as the resolution → claim goes to **Submitted for QA**.
2. QA Reviewer (or Team Lead/RCM Manager) approves or rejects.
3. On approval, the claim appears in an **Approved Write-Offs – Pending Posting** list in the same agent’s queue, with a notification: “Write-off approved — perform the write-off in the PMS.”
4. Agent posts the write-off in the PMS and marks the claim complete.
5. If the claim is still open on the next sync, notify the agent and AR Manager (Section 5).

Rationale: prevents bulk write-offs of aged claims used to reduce queue balances, while keeping posting with the agent since the system is not PMS-integrated.

8.4 Internal escalation and reassignment requests

- **Escalate to Supervisor** — internal only (e.g., payer portal access required but unavailable). Routed to both Team Lead and AR Manager; either can respond.
- **Request Reassignment** — agent returns claims they cannot work (e.g., requires a payer call outside their shift) for supervisor reassignment.
- **New:** add a mandatory **reason category** dropdown to both, alongside the free-text box, so escalation and reassignment volumes can be analysed by reason.

8.5 Batch assignment due date

- The Due Date on batch assignment is **optional**. Use it for urgent batches, e.g., claims approaching timely filing limit (TFL), where a short deadline (e.g., 2 days) applies to every claim in the batch.

9. Client Escalation (CIP) Workflow

9.1 Flow

1. Agent selects resolution **CIP (Client Escalation)**. Two comment fields are required: the internal **Follow-Up Comment** and a client-facing **CIP Comment** (professional wording). The CIP Comment box appears only when CIP is selected.
2. Claim goes to **Submitted for QA**. QA can be performed by a QA Reviewer, Team Lead or RCM Manager.
3. After QA approval, the claim enters the **CIP Escalations queue – Pending Approval**, accessible only to Team Lead, RCM Manager and System Administrator. They approve individually or in bulk (second approval layer, as CIPs are account-management level communications).
4. Approved CIPs appear in the client's **Escalation Requests** tab.
5. Client responds with the requested information and may attach files; status changes to **Client Responded** and the claim returns to the CIP Escalations queue.
6. Manager/TL reviews (daily, weekly or bi-weekly as preferred). If the response is insufficient, they add a note and **re-escalate** to the client. If sufficient, they approve and the claim is **automatically returned to the original agent**.

9.2 Comment visibility

- Internal follow-up comments and client-facing CIP comments must always be stored and displayed **separately**.
- For escalated claims, client users see only the CIP comment and client response — never internal notes.
- Client access to work-queue/follow-up data is an **access-level setting**. By default clients see Escalations, Analytics and Reports only. If a client is granted follow-up visibility (e.g., an “All Claims” tab), they may see follow-up notes on non-CIP claims, still separated from CIP communication.

10. Roles, Access & User Management

Capability	Sys Admin	AR/RCM Mgr	Team Lead	QA Reviewer	AR Agent	Client User
View overall / restricted AR queues	✓	✓	✓			
Assign / reassign claims	✓	✓	✓			
Work assigned claims					✓	
QA approve / reject	✓	✓	✓	✓		
Approve CIP to client	✓	✓	✓			
Respond to CIP						✓
Maintain NC & Auto-Adjust lists	✓	✓				
Export Auto Adjustments report	✓	✓	✓			
View follow-up notes	✓	✓	✓	✓	✓	If granted
Analytics & reports	✓	✓	✓			✓ (scoped)

10.1 Client-level user provisioning

- Admin creates client users in User Management and scopes them by **Lab → Clinic**.
- Cascading selection: selecting a Lab loads only that lab's clinics (and referring entities); selecting a clinic restricts the user to that clinic's data.
- CIP escalations for a clinic are routed to that clinic's scoped user(s).

11. New Features & Enhancements Summary

#	Feature	Raised by	Phase
1	Full master sync with rule-based AR queue classification	Banuka	Phase 1
2	Non-Collectible denial list in System Settings	Banuka / Prasanna	Phase 1
3	Assigned claims deducted from denial-insight counts	Banuka	Phase 1
4	Expandable CPT line-level detail with line-level denials	Banuka	Phase 1
5	CARC prefix normalization	Banuka	Phase 1
6	Revenue Expectation, Potential Recovery & Recovery Status logic	Banuka / Prasanna	Phase 1
7	Re-Follow-Up split: Unresolved Claims / New Denials	Banuka / Azwin	Phase 1
8	Auto Adjustments queue + PMS bulk-posting export	Banuka	Phase 1
9	Re-appearing adjusted/written-off claims detection and notification	Prasanna / Banuka	Phase 1
10	Write-off QA approval with pending-posting list	Prasanna	Phase 1
11	Attachments on all follow-ups, categorized, with Document Vault (Azure Blob, HIPAA)	Prasanna / Banuka	Phase 1
12	Reason categories for escalation and reassignment requests	Prasanna	Phase 1
13	Optional batch due date	Prasanna	Phase 1
14	Clinic-scoped client users (Lab → Clinic cascade)	Banuka	Phase 1
15	Denial-hierarchy mapping to derive primary denial	Banuka	Phase 2 (on mapping delivery)
16	Action-based follow-up timelines (e.g., rebill 30d, appeal 60d)	Prasanna / Azwin	Phase 2
17	Basic forecasting model (manual model currently in use)	Banuka	Phase 1 – basic; payer-policy model later
18	Payer-policy-driven prediction (Payer Policy Database)	Client request	Future — after AR workflow is complete

12. Delivery Plan & Priorities

- **Timeline:** build starts 29 September 2026; target completion **13 November 2026**. Leadership (William) was committed to October; the October–November window will be positioned as a marketing/demo phase using the existing HTML demo, with ~30 days set as the onboarding expectation for new clients.
- **Approach:** build the common logic framework once, then plug in client data one client at a time. Each module is handed over for testing at the end of its build — no big-bang testing at the end.
- **Reuse:** most existing Denial Workflow components are reusable; the principal new work is the queue logic.
- **First client:** build and test with **InHealth** (not Co), so the GSP team can test on live work.
- **Priority accounts:** Dan’s and Dakota’s clients (active AR engagements) — reporting module and denial workflow. Dakota accounts named: Beach Tree, InHealth, Rizan Tide (spelling as transcribed — verify). Data validation this week; target to confirm onboarding by next Wednesday.
- **Reports:** one complex ongoing client report (prepared by Banuka for Dakota’s clients) needs a 30-minute walkthrough session; other reports are straightforward.
- **Forecasting:** implement the current manual forecasting model (training via Azwin/Azeem); not connected to the Payer Policy Database in this phase.

13. Open Items & Decisions Pending

#	Item	Owner	Needed by
1	Denial-hierarchy mapping (primary/latest denial derivation)	Banuka	Phase 2 start
2	Denial categorization for all denial codes	Banuka	Before queue engine build
3	Initial Non-Collectible and Auto-Adjust code lists	Banuka	Before queue engine build
4	Confirm restricted (“hidden”) queue set and evaluation precedence	Banuka	Before queue engine build
5	Revenue Expectation rate source (Medicare fee schedule vs. historical allowed) and fallback	Banuka	Before financial fields build
6	Definition of “No Recovery” status	Banuka	Before financial fields build
7	Final attachment size limit (10 vs. 15 MB) and document categories	Prasanna / Azwin	Before attachments build
8	Next follow-up date override rules	Banuka	Before follow-up build
9	Escalation and reassignment reason category list	Banuka / Azwin	Before follow-up build
10	Walkthrough of the complex client report	Banuka → Prasanna	Next session
11	Forecasting model training	Azwin / Azeem	TBD

14. Suggested Acceptance Tests

#	Scenario	Expected result
1	Claim with Ins Bal = 0, Pt Bal = 0, payment received	Closed – Fully Paid; not in work queue
2	Ins Bal > 0, no payment/denial/adjustment	Non-Responder; auto-fed to work queue
3	Ins Bal > 0, primary denial 197 (on NC List)	Denied – Non-Collectible; visible only to Admin/Mgr/TL
4	Claim with “PR 204” and “CO 204” on separate claims	Both map to code 204 and the same category
5	Remark code N57 on a claim	Stored and mapped as N57 (letter retained)
6	Pt Bal > 0, Ins Bal = 0, denial present	Patient AR (denial ignored)
7	Insight shows 50 claims; manager assigns 20	Insight count/balance reflect 30 remaining
8	Revenue Expectation \$353; ins payment \$120; pt balance \$30	Potential Recovery \$203; Recovery Status Partially Recovered
9	Completed claim receives new denial on next sync	Moves to Re-Follow-Up – New Denials, unassigned, history intact
10	Assigned, untouched claim’s new denial is on NC List	Auto-processed and removed from agent queue
11	Agent submits write-off; QA approves	Appears in agent’s pending-posting list with notification
12	Auto-adjusted claim still open on next sync	Flagged, returned to Auto Adjustments, agent & manager notified
13	CIP escalated, client responds, manager approves	Claim returns automatically to original agent
14	Client user scoped to Clinic A	Sees only Clinic A data and CIPs; never internal notes on CIPs
15	Agent saves follow-up today	Next follow-up date defaults to today + 45 calendar days
16	Upload a 20 MB file	Rejected with size-limit message

PART B — DETAILED FUNCTIONAL REQUIREMENTS

Source codes used in the tables: **M** = mockup behaviour (LRN_Denial_AR_Workbench_Demo_Account.html), **BR** = AR Workbench Build Reference, **S** = session decision (Part A), **N** = new requirement introduced to close a gap. Where a requirement is marked **M→S**, the mockup behaviour is modified by a session decision.

15. Roles, Permissions & Data Scoping (FR-ACC)

15.1 Roles

Role key	Display name	Summary
admin	System Administrator	Everything: assign, edit claims, QA decisions, approvals, user management, audit, settings, client management, all clients.
manager	RCM Manager (AR Manager)	Assign, edit claims, QA decisions, approvals (CIP), audit, data processing, all clients. No user management or settings.
lead	Senior AR Analyst / Team Lead	Assign, edit claims, QA decisions, CIP approvals and request responses, all clients. No audit, settings or user management.
agent	AR Agent	Edit claims in own caseload only (log follow-ups, escalate, request reassignment).
qa	QA Reviewer	QA decisions, all clients. Read access to claims for review.
viewer	Client / Management Viewer	Read-only dashboards, analytics, reports; responds to CIP escalation requests for own scope.

15.2 Permission matrix (server-enforced)

Permission	admin	manager	lead	agent	qa	viewer
assign (assign/reassign, bulk assign, auto-adjust)	✓	✓	✓			
editClaim (log follow-up)	✓	✓	✓	✓ own		
qaDecide	✓	✓	✓		✓	
approve (CIP to client)	✓	✓	✓ (S)			
manageUsers	✓					
viewAudit	✓	✓				
manageSettings	✓					
allClients	✓	✓	✓		✓	
viewClientMgmt	✓					

Note: in the mockup the lead has approve = false in the permission table but can still act in the CIP queue via the assign permission. Session decision: Team Lead **can** approve CIPs — grant approve to lead explicitly.

15.3 Requirements

ID	Requirement	Source
FR-ACC-01	Every user has one role and one access grant. Access grant levels: All Clients , Client (one lab), Clinic (one ordering practice within a client) or Referring Provider (one provider within a client).	BR
FR-ACC-02	All claim reads — every screen, count, badge, export, report and API — must pass through one central scoping function: filter by allowed client(s) → narrow by clinic or provider when the grant requires → narrow to assignedAgent for the Agent role → apply the header client selector.	BR
FR-ACC-03	Users with a Client/Clinic/Provider grant see a fixed scope badge (e.g. “Summit Toxicology Labs → Northgate Orthopedic Associates (Clinic)”) in place of the All Clients dropdown. Unrestricted users get an All Clients / per-client selector in the header.	M
FR-ACC-04	Role gates which navigation items and actions exist; access grant narrows data within them. Route list and allowed roles are declared once and used for sidebar, routing and API authorization (no drift between UI and enforcement).	BR
FR-ACC-05	An AR Agent sees only claims assigned to their agent code in every view (Work Queue, My Work, Follow-Up, Analytics, Reports).	M
FR-ACC-06	Session rule (S): client users never see internal follow-up notes on escalated claims; access to follow-up notes on non-CIP claims is an optional, admin-granted capability.	S

16. Navigation Map & Notification Badges (FR-NAV)

Screen	Roles	Badge count (live)
Dashboard	All	—
Client Management	admin	—
Data Processing	admin, manager	—
Denial & AR Work Queue	admin, manager, lead, agent	Unassigned claims
Assignment Management	admin, manager, lead	—
My Work	agent, lead	Open (not financially closed) claims in caseload
Follow-Up Management	admin, manager, lead, agent	Follow-up-actionable claims (same rule as the screen)
QA Verification Queue	admin, manager, lead, qa	Submitted for QA
CIP Escalations	admin, manager, lead	CIP cases Pending Approval + Client Responded
Escalation & Reassignment Requests	admin, manager, lead	Pending requests
Escalation Requests (client portal)	viewer	Cases Sent to Client (awaiting their response)
Recovery & Financial Analytics	All	—
Reports	All	—
User Management	admin	—

Screen	Roles	Badge count (live)
Audit Logs	admin, manager	—
System Settings	admin	—
Document Vault (N)	admin, manager, lead, agent, qa	—

ID	Requirement	Source
FR-NAV-01	After login, land on the first navigation item permitted for the role.	M
FR-NAV-02	Badge counts must use exactly the same predicate as the destination screen's base filter so the badge and the list never disagree.	M
FR-NAV-03	Header shows data refresh date and source period (e.g. "Data refreshed Sep 15, 2026 · Source period Jan 01, 2025 – Sep 08, 2026").	M
FR-NAV-04	Claim Detail is reachable from every queue; its Back button returns to the originating screen (Work Queue, My Work, QA, Audit, CIP Escalations, Requests, Follow-Up, Assignment).	M

17. Data Processing & Denial Analysis Insights (FR-DP)

ID	Requirement	Source
FR-DP-01	Data Processing screen shows last refresh date, source period start/end, and record count, with a Run Data Processing action (admin, manager).	M
FR-DP-02	Run Data Processing executes: (1) sync weekly claim-level and line-level master files for each active lab; (2) reconcile; (3) normalize denial codes; (4) derive primary denial, category, financial fields; (5) apply re-sync rules (Section 5 / FR-SYNC); (6) re-classify AR queues; (7) load current and previous week denial insights. Progress steps are displayed; the run is logged with start/end, row counts and errors.	M→S
FR-DP-03	Denial Analysis Report — Insights table columns: #, Denial Code, Description, # of Denials (claims), Total Balance (\$), Highest \$ Impact Insurance, Ins. Balance (\$) for that payer, \$ Impact (%), Observation, Category tag (Review / Appeal-MR), Recommended Action, Route to Work Queue .	M
FR-DP-04	Insight rows are grouped by primary denial code (S) — the mockup splits comma-separated codes and counts a claim under every code, which must not be carried over. Only open claims (not financially closed) are counted, and (S) assigned claims are excluded so the row shows the unassigned remainder.	M→S
FR-DP-05	Highest \$ impact payer = payer with the largest share of the code's outstanding balance; \$ Impact % = that payer's balance ÷ code balance. Counts and balances cover all payers.	M
FR-DP-06	Observation text: "Per review, the majority of denied claims are for {most common CPT/service line}." Recommended action and category tag come from a Denial Category → Action mapping (admin-maintainable; seeded from the mockup's 11-category table).	M
FR-DP-07	Route to Work Queue opens the Work Queue filtered to that insight's claim IDs, with a removable "From insight: Denial Code X" chip.	M
FR-DP-08	Dashboard shows the top 5 insights (current week); Data Processing shows the full list (current + previous week, unresolved only). Viewers do not see insights.	M→S

18. AR Queue Classification Engine — Final Specification (FR-ARQ)

This section merges the mockup's `classifyQueue()` precedence with the session decisions. It is the single definition developers should implement (config-driven, re-evaluated on every sync and every workflow action).

18.1 Lifecycle predicates

Predicate	Definition
isFinanciallyClosed	Insurance balance ≤ 0 and patient balance ≤ 0 (tolerance \$0.005).
isWorkComplete	isFinanciallyClosed OR workflowStatus = Completed.
isOpenInsuranceAR	Insurance balance > 0 . Drives every agent-facing inventory view.
daysSinceLastTouch	Days since the most recent activity-log entry of any kind (including intake "Claim Identified"). Used for the Unassigned-Claims staleness view.
refollowupDue	Only for claims worked at least once (lastFollowUp not null): days since lastFollowUp ≥ 45 , OR nextFollowUp date $\leq$ today.
isFollowUpActionable	(isOpenInsuranceAR OR adHocFollowUpAssigned) AND workflowStatus $\in$ {Assigned, QA Rejected}.
isNonCollectible (S)	Normalized primary denial code is in the Non-Collectible master (mockup checks any code on the claim — do not copy).

18.2 Workflow statuses

Unassigned $\rightarrow$ Assigned $\rightarrow$ Submitted for QA $\rightarrow$ (QA Rejected $\leftrightarrow$ Assigned rework) $\rightarrow$ Completed. Logging a follow-up always moves the claim to Submitted for QA. Assigning an Unassigned or Completed claim sets Assigned. Returning a CIP case to the agent re-opens the claim to Assigned.

18.3 Precedence (evaluate top to bottom, first match wins)

#	Condition	Queue / sub-queue
1	QA record status = Awaiting QA	Submitted for QA
2	workflowStatus = QA Rejected	QA Rejected (auto-assigned to same agent)
3	(N) Balance nullified by Auto-Adjust or approved Write-Off and not yet confirmed posted in PMS	Auto Adjustments $\rightarrow$ System Auto-Adjusted / Approved Write-Offs (sub-queues — to confirm)
4	workflowStatus = Completed AND insurance balance ≤ 0	Fall through to financial state (#8)
5	workflowStatus = Completed AND an open CIP case exists (Pending Approval / Sent to Client / Client Responded)	Client Escalations Pending
6	workflowStatus = Completed AND (S) new denial received on sync	Re-Follow-Up Required $\rightarrow$ New Denials
7	workflowStatus = Completed AND refollowupDue	Re-Follow-Up Required $\rightarrow$ Unresolved Claims

#	Condition	Queue / sub-queue
7a	workflowStatus = Completed (otherwise)	Completed Claims
7b	A CIP case was Returned to Agent with a client response and the claim is back in Assigned	CIP Response Received
8	Financial state: Ins Bal $\leq$ 0 and Pt Bal $>$ 0	Patient AR
8a	Ins Bal $\leq$ 0, Pt Bal $\leq$ 0, payments received	Closed $\rightarrow$ Fully Paid
8b	Ins Bal $\leq$ 0, Pt Bal $\leq$ 0, no payment	Closed $\rightarrow$ Fully Adjusted
8c	Ins Bal $>$ 0 and insurance payment $>$ 0	Partially Paid $\rightarrow$ Possible Collectible / Non-Collectible
8d	Ins Bal $>$ 0, no payment, denial present	Denied $\rightarrow$ Possible Collectible / Non-Collectible
8e	Ins Bal $>$ 0, no payment, no denial, adjustment $>$ 0	Partially Adjusted $\rightarrow$ Possible Collectible
8f	Ins Bal $>$ 0, no payment, no denial, no adjustment	Non-Responded AR $\rightarrow$ Possible Collectible

Differences from the mockup: The mockup classifies Non-Responded only when denialCategory = “Unresponsive Payers”, treats any adjustment without payment as Partially Adjusted even when a denial exists, checks all codes for non-collectibility, and has no Auto Adjustments or New Denials queues. Implement the table above.

ID	Requirement	Source
FR-ARQ-01	Queue taxonomy (labels, sub-queues, priority flag) is master data, not code. Priority leaves are those needing agent work; non-priority leaves are Closed, Patient AR and Non-Collectible sub-queues.	M
FR-ARQ-02	Every claim shows its AR queue as a badge “Top queue · Sub-queue” on lists and Claim Detail.	M
FR-ARQ-03	Classification is recomputed after every sync and every state-changing action; a nightly snapshot of queue membership is stored for trend/movement reporting (plan T038).	M $\rightarrow$ N
FR-ARQ-04	Non-priority leaves (Closed, Patient AR, Non-Collectible) are visible only to admin, manager and lead; they can assign such claims ad hoc (flagged adHocFollowUpAssigned, audit “Ad-Hoc Assignment”).	M

19. Dashboard (FR-DASH)

19.1 KPI tiles

KPI	Definition
Total Eligible Claims	Count of scoped claims; delta chip = claims identified this week vs last week (submission date in last 0–7 vs 8–14 days).
Total Outstanding Insurance AR	Σ insurance balance; “as of {refresh date}”.
Claims Awaiting Assignment	workflowStatus = Unassigned; hint “needs manager action” when $>$ 0.

KPI	Definition
Claims In Progress	workflowStatus = Assigned.
Claims Awaiting QA	workflowStatus = Submitted for QA.
QA Rejected	workflowStatus = QA Rejected.
Completed / Closed Claims	isWorkComplete.
Total Recovered Amount	Σ recovered amount (S: insurance payments dated after the denial date).
Potential Recovery Amount	Σ Potential Recovery (S formula, Section 7) for claims not financially closed.
Overdue Follow-Ups	nextFollowUp < today and not financially closed.

ID	Requirement	Source
FR-DASH-01	Widgets: Latest Denial Analysis Report insights (top 5, non-viewers); AR Collections Progress (expected revenue per AR queue with link to full report); Claim Queue Volumes (admin/manager/lead; workable leaves with open insurance balance, excludes Closed & Patient AR); Denial Category Distribution (by \$ balance); AR Aging Distribution (count by aging bucket, sequential colour ramp); Claim Workflow Status; Recovery Performance donut (recovered vs outstanding, % = recovered $\div$ initial insurance AR); Agent Productivity table (non-viewers).	M
FR-DASH-02	Session control plan (project brief): for each denial classification AND each action/task bucket show claim count and outstanding \$ split into Open, In Progress and Closed buckets. Mockup does not have this — add as a Control Plan widget with a Classification / Action toggle.	N
FR-DASH-03	SLA metrics (project brief): SLA attainment for Appeals, Rebills, Write-offs; Appeal Success Rate; AR Aging. Mockup does not have these — add tiles once TAT per action is configured (plan D-11).	N
FR-DASH-04	Every widget respects scoping; clicking a queue volume bar opens the Work Queue filtered to that queue.	M $\rightarrow$ N

20. Denial & AR Work Queue (FR-WQ)

ID	Requirement	Source
FR-WQ-01	Base population: agents see only open-insurance-AR claims in priority queues; admin/manager/lead see the full register including Closed, Patient AR and Non-Collectible.	M
FR-WQ-02	Primary filters (multi-select): Search (claim ID, patient account, provider), Client, Payer, Panel Type, Denial Category, Workflow Status, Assigned Agent, Priority, AR Aging, AR Queue; TFL-risk-only toggle.	M
FR-WQ-03	More filters: Claim Status (ingested: Fully Denied, Partially Paid, No Response, Partially Denied...), Rendering Provider, Fix/Resolution, CPT code (contains), Insurance Balance $\geq$ / $\leq$, Next Follow-Up From / To.	M
FR-WQ-04	Actions: Clear Filters, Save View (named, per user per queue — server-persisted), apply/delete saved view chips, insight chip removal.	M

ID	Requirement	Source
FR-WQ-05	Columns: Claim ID (link), Client, Patient Acct, DOS, Payer, Panel Type, CPT (first + “+n”), Denial Category, Denial Reason*, Claim Status*, Ins. Balance, Aging, TFL (At Risk / OK), Priority, Assigned Agent, Workflow Status, AR Queue, Fix/Resolution*, Last Follow-Up, Next Follow-Up (red if overdue, amber if ≤ 2 days), Open. (* hidden by default)	M
FR-WQ-06	Bulk assign (assign permission): row checkboxes → Assign Selected → modal with agent picker and optional note; already-assigned claims are reassigned and the previous agent is recorded in audit.	M
FR-WQ-07	Process Automatic Adjustments button (admin/manager/lead): see FR-ADJ.	M→S
FR-WQ-08	S: claim listing is claim-level; row count reflects claims, not lines.	S

21. Claim Detail Workspace (FR-CD)

ID	Requirement	Source
FR-CD-01	Header: Claim ID, workflow status badge, AR queue badge, TFL At Risk badge, pending Escalation/Reassignment request badges; identifiers — client, payer, panel, DOS, patient account, rendering provider, ordering practice.	M
FR-CD-02	Four-stage stepper: Identified → Assigned → QA Verification → Completed (QA Rejected shows at QA stage). Also show the category workflow template path (e.g. Coding: Identified → Assigned → Coding Review → Correction Required → Corrected Claim Submitted → Payer Follow-Up → Payment/Resolution → QA → Closed). Five templates: Documentation Required, Non-Responded AR, Coding/Claims Edit, Partially Paid, Denied.	M
FR-CD-03	Tabs (shown when relevant): Overview, CPT/Line Detail, Denial Info, Follow-Ups (count), Activity Timeline, QA Review (when a QA record exists), CIP Case(s) (when a case exists; count if > 1), Documents (N).	M→N
FR-CD-04	Overview stat strip: Billed, Allowed, Insurance Payment, Patient Payment, Adjustments, Insurance Balance, Patient Balance, Revenue Expectation (S rename), Actual Payment, Variance, Potential Recovery (S formula), Recovery Status (S values). Payment-breakdown bar (Insurance Paid, Patient Paid, Adjustments, Outstanding). Underpayment banner when underpayment > 0 (“Appeal recommended” when appealRequired).	M→S
FR-CD-05	CPT/Line Detail columns: CPT, Description, Diagnosis, Units, Billed, Allowed, Paid, Adjusted, Denied, Balance, Status. (S) add an expand control per line to show line-level denial codes (original prefix kept for display), descriptions and classifications.	M→S
FR-CD-06	Denial Info: Denial Category, Primary Denial Code, Denial Reason, Denial Date, Latest Follow-Up (fix · status), AR Queue, Corrective Action Required (by category), Appeal Deadline (TFL date; “at risk” in red), Supporting Documentation Required (Yes when category = Additional Documentation Required).	M
FR-CD-07	Follow-Ups tab: AR queue and next follow-up date summary; reverse-chronological note cards with all Comments Framework fields; CIP block rendered separately inside the note.	M
FR-CD-08	Activity Timeline: reverse-chronological audit entries with user, role, action, description and previous → new value.	M
FR-CD-09	Role-gated header actions: Assign/Reassign (assign permission); Log Follow-Up Note (editClaim, not pending QA, not financially closed); Escalate to Supervisor and Request Reassignment (agent role only).	M

22. Log Follow-Up Note — Comments Framework (FR-FU)

Field	Rule	Values (seed)
Claim Type	Required	Primary, Secondary
Follow Up Type	Required	Review, Online, Call
Claim Status	Required; defaults to status on the last note, else In Process	Paid, Denied, Not Received, In Process, Paid to Patient, Patient Responsibility, On Hold
Denial Root Cause	Shown and required only when Claim Status = Denied; stored null otherwise	15 values (master: Denial Root Cause Options)
Fix / Resolution	Required; option list filtered by Claim Status (Fix-by-Status master)	e.g. Denied → Appealed, Billed to Patient, Billed to Secondary, CIP - Client Escalations, Medical Records Submitted, Reconsideration Submitted, Reprocessed, Resubmitted - E/F/P, Write Off
Follow Up Comment	Required, free text (internal)	—
Next Follow-Up Date	(S) default today + 45 calendar days (mockup uses +7 — do not copy)	—
CIP Category	Only when Fix = CIP - Client Escalations	DX Codes, Insurance, Enrollment, Documents
Required Information	Only when CIP	11 values (e.g. Additional Documentation Required – Chart/History Notes; Active Current Insurance info required; Valid ICD-10 Codes Required)
CIP Comment	Required when CIP; client-facing, professional wording	—
Attachments (N)	Optional, multiple; category required per file; size limit per FR-DOC	—

ID	Requirement	Source
FR-FU-01	Saving a note: appends the follow-up record; sets lastFollowUp, fixResolution, denialRootCause, nextFollowUp, workedStatus = Worked; creates a QA record (Awaiting QA, isEscalation flag when CIP); sets workflowStatus = Submitted for QA; writes “Follow-Up Logged” and “Submitted for QA” audit entries. There is no separate Submit-for-QA step.	M
FR-FU-02	The ingested Claim Status (Fully Denied / Partially Paid / No Response ...) is a different field from the Comments Framework Claim Status and must not be overwritten.	M
FR-FU-03	(S) CIP: the CIP case is created on save but becomes visible in the CIP Escalations “Pending Approval” stage only after QA approves the note (mockup creates it immediately in parallel with QA).	M→S
FR-FU-04	(S) Write Off selected as Fix/Resolution follows the write-off approval loop in FR-ADJ.	S
FR-FU-05	Log Follow-Up is also available from My Work and Follow-Up Management rows.	M
FR-FU-06	Bulk follow-up update via Excel upload on the same field set (plan T030) — see FR-XL.	N

23. Assignment Management (FR-ASG)

ID	Requirement	Source
FR-ASG-01	Three sections drawing from different populations. Unassigned Claims: workflowStatus Unassigned, open insurance AR, daysSinceLastTouch ≥ 45; columns Claim ID, Client, Payer, Panel, Denial Category, Ins. Balance, Aging, Priority, Days Untouched; select → Assign Selected.	BR
FR-ASG-02	Create Assignment Batch: criteria Client, Denial Category, Payer, Panel Type, Priority (multi-select) over all unassigned open-AR claims (no 45-day restriction); live preview “n claims · \$x total insurance AR”; agent picker; Due Date optional (S); Create Batch & Assign disabled until criteria match ≥ 1 claim. Session: batches may also be created by Action/Task (N — add Action filter).	M→S
FR-ASG-03	Assignment Batches log: Batch ID (BATCH-nnnn), Client, Category, # Claims, Total Ins. AR, Assigned Agent, Created By, Assign Date, Due Date, Status, Completion % (work-complete ÷ claims). Server-persisted (mockup is session-only).	M
FR-ASG-04	Agent Workload chart: open claims per agent = assigned, not work-complete, open insurance AR.	M
FR-ASG-05	Assigned Claims — Bulk Reassign: every assigned claim with open insurance AR (includes Completed-but-unpaid); filters Client, Current Agent, Denial Category, Panel, Priority; Reassign Selected.	BR
FR-ASG-06	Reassigning a claim auto-resolves any pending Reassignment Request on it (“Resolved by reassigning to X”).	M
FR-ASG-07	(S) Assigning claims removes them from the denial-insight outstanding counts.	S

24. My Work (FR-MW)

ID	Requirement	Source
FR-MW-01	Scope: agent = own assigned claims; lead = the whole team’s assigned claims.	M
FR-MW-02	Quick-filter KPI tiles (click to filter): Total Assigned, Due Today, Overdue, High Priority, Re-follow-up Required, CIP Response Received, Awaiting Payer Response (last fix ∈ Pending Payer Adjudication, Resubmitted E/F/P, Reconsideration Submitted, Appealed), Submitted for QA, Rejected QA Items. (N) add Approved Write-Offs – Pending Posting and New Denials tiles.	M→N
FR-MW-03	Filters: Search, Payer, Panel, Denial Category, Priority, Status, Claim Status, Fix/Resolution, Assigned Agent, TFL-risk-only; saved views.	M
FR-MW-04	Columns include Denial Category, Ins. Balance, Priority, Status, Fix/Resolution*, AR Queue, Next Follow-Up, and a Log Follow-Up action (hidden when pending QA or financially closed).	M

25. Follow-Up Management (FR-FUM)

ID	Requirement	Source
FR-FUM-01	A to-do queue: base population = isFollowUpActionable. A claim drops off the moment a follow-up is logged (status becomes Submitted for QA).	M

ID	Requirement	Source
FR-FUM-02	Window tiles: All Active, Overdue, Due Today, Upcoming, No Follow-Up Scheduled.	M
FR-FUM-03	Filters: Search, Client, Payer, Panel, Denial Category, Priority, AR Aging, Assigned Agent, Workflow Status, Claim Status, Fix/Resolution, TFL-risk-only; saved views.	M
FR-FUM-04	Next Follow-Up column shows “(n d overdue)” in red or “(today)” in amber; Log Follow-Up action per row.	M
FR-FUM-05	Re-follow-up Required logic (worked claims): 45+ days since last follow-up or next follow-up date reached — separate from the Unassigned staleness rule.	BR

26. QA Verification Queue (FR-QA)

ID	Requirement	Source
FR-QA-01	Population: every claim with a QA record. Filters: Search (claim, agent), Client, Payer, Panel, Denial Category, Agent, QA Reviewer, Escalation (Yes/No), QA Status.	M
FR-QA-02	Columns: Claim ID, Client, Payer, Panel, Agent, Denial Category, Ins. Balance, Agent Completed date, QA Status, Escalation (CIP badge), QA Reviewer, Error Type, Action (Approve / Reject, else Review).	M
FR-QA-03	Self-approval block: a user can never approve their own submission. (Mockup only checks the agent role — enforce for all roles.)	M→N
FR-QA-04	Approve → QA Approved, workflowStatus Completed. Reject → requires Error Type (Incomplete Documentation, Incorrect Denial Category, Missed Follow-Up, Financial Update Error, Other) and a note; status QA Rejected; auto-returns to the same agent.	M
FR-QA-05	QA Review tab shows six scoring criteria: Correct claim analysis, Correct denial categorization, Appropriate action taken, Proper documentation, Accurate financial updates, Correct/timely follow-up.	M
FR-QA-06	Bulk approve: only eligible rows get a checkbox; confirmation shows count, total insurance AR and number of CIP escalations; result toast reports approved and skipped (own work / already decided).	BR
FR-QA-07	(S) QA approval of a CIP note releases the CIP case to Pending Approval; QA approval of a Write-Off note routes to Approved Write-Offs – Pending Posting.	S

27. Escalation & Reassignment Requests (FR-REQ)

ID	Requirement	Source
FR-REQ-01	Agent actions from Claim Detail: Escalate to Supervisor and Request Reassignment , each with a required reason text and (S) a required reason category .	M→S
FR-REQ-02	Each request record: id, type, note, reason category (N), requested by/role/date, status Pending/Resolved, resolved by/role/date, resolution note.	M
FR-REQ-03	Queue (admin, manager, lead) flattens requests across claims, pending first. Filters: Search, Client, Payer, Request Type, AR Agent, Requested By, AR Queue, Status (+ Reason Category N). Columns: Claim ID, Client, Payer, AR Agent, Request Type, Reason/Note, Ins. Balance, Current AR Queue, Requested By, Requested, Status, Response, Action.	M

ID	Requirement	Source
FR-REQ-04	Respond modal: Mark Resolved (note required for escalations), Open Claim, and for reassignment requests Reassign & Resolve .	M
FR-REQ-05	Bulk resolve: only pending rows selectable; shared resolution note required; summary shows escalation vs reassignment counts.	BR
FR-REQ-06	Escalate to Supervisor in the mockup also moves the claim to Submitted for QA. Session described it as going to TL and Manager. To confirm whether the claim should stay with the agent (recommended) or enter QA.	M→?

28. CIP Escalations — Internal Queue (FR-CIP)

28.1 CIP case lifecycle

State	Entered when	Visible to client?
Pending Approval	(S) QA approves an agent note with Fix = CIP - Client Escalations	No
Sent to Client	TL / RCM Manager / Admin approves (single or bulk); also when a response is marked insufficient (round + 1)	Yes
Client Responded	Client submits response text and attachments (individual or bulk CSV)	Yes
Returned to Agent	Response accepted, or request rejected before sending; claim re-opens to Assigned	Yes (resolved)

ID	Requirement	Source
FR-CIP-01	Case record: id, category, required information, CIP comment, logged date, requested by/role, status, round, client response {text, attachments, responder, date}, last review decision {approved / insufficient / rejected, by, date, note}, history[] of every transition (actor, role, action, note).	M
FR-CIP-02	Queue header shows live counts per stage. Columns: Claim ID, Client, Payer, Status (with “Round n” when > 1), CIP Category, Required Information, AR Agent Request, Ins. Balance, Current AR Queue, Requested By, Logged Date, Client Response, Attachments (download links), Action.	M
FR-CIP-03	Row actions: Pending Approval → Review & Approve (approve, or reject back to agent with reason); Client Responded → Review Response (approve & return to agent, or mark insufficient with a reason visible to the client).	M
FR-CIP-04	Bulk Approve Selected / Bulk Reject – Send Back Selected: mixed selections allowed; each row gets the action for its own stage; Sent to Client / Returned rows are skipped; one shared note is recorded and tagged “Bulk action” in each case history.	BR
FR-CIP-05	Default sort: Client Responded, Pending Approval, Sent to Client, Returned to Agent; then newest.	M
FR-CIP-06	Claim Detail CIP tab is read-only with a “Manage in CIP Escalations Queue” link for managers.	M
FR-CIP-07	The CIP queue is exportable so account managers can share weekly/bi-weekly with clients.	M→S

29. Client Portal — Escalation Requests (FR-CLI)

ID	Requirement	Source
FR-CLI-01	Visible only to viewer users within their scope; never shows Pending Approval cases.	M
FR-CLI-02	Tiles: Awaiting Your Response (Sent to Client), Submitted – Under Review (Client Responded), Resolved (Returned to Agent).	M
FR-CLI-03	Columns: Claim ID, Payer, Patient Acct, DOS, Status (Round n), Category, Information Requested, Request From Your AR Team, Reviewer Feedback (“Needs more info” + reason when marked insufficient), Action (Respond / View).	M
FR-CLI-04	Respond modal: response text (required) + attachments. Mockup limit is 3 files × 250 KB (demo only); production limit per FR-DOC (10–15 MB).	M→S
FR-CLI-05	Bulk response: Download Template (CSV: Case ID, Claim ID, Patient Acct, DOS, Rendering Provider, Payer, CIP Category, Required Information, Request From AR Team, Response, Attachment Notes) → Upload Completed Template. Validation result modal: updated, skipped (blank response), skipped (not awaiting), skipped (case not found), duplicates ignored.	M
FR-CLI-06	Client users also have Dashboard, Analytics and Reports (scoped); Escalations, Analytics and Reports are the default access set (S).	S

30. Automatic Adjustments & Write-Offs (FR-ADJ)

ID	Requirement	Source
FR-ADJ-01	Eligible for auto-adjustment: claims with open insurance balance whose primary denial code is on the Auto-Adjust list (S: e.g. 242, 197) or Non-Collectible list (mockup). To confirm whether the two lists are the same master or separate.	M→S
FR-ADJ-02	Trigger: Process Automatic Adjustments (Work Queue / Data Processing; admin, manager, lead). Confirmation shows claim count and total \$; single-claim and batch supported.	M
FR-ADJ-03	Effect: add adjustment = remaining balance, set remaining insurance AR to 0 in the Workbench, add an automatic system comment, write an audit entry with actor “System / Automation”, and place the claim in the Auto Adjustments queue (S) — not Completed/Closed as in the mockup.	M→S
FR-ADJ-04	Export: AR Manager/TL can export the Auto Adjustments queue (claim, patient, DOS, payer, CPT lines, amount, reason code) for bulk posting in the PMS; a “Mark as Posted” bulk action closes the items after posting (N).	S→N
FR-ADJ-05	Write-off loop (S): agent Fix = Write Off → QA → on approval claim appears in agent’s Approved Write-Offs – Pending Posting list with notification → agent posts in PMS and marks complete.	S
FR-ADJ-06	Re-appearance (S): if an adjusted/written-off claim still carries a balance on the next sync, flag “Adjustment not posted”, return it to the Auto Adjustments queue and notify the agent and AR Manager.	S

31. Recovery & Financial Analytics (FR-AN)

ID	Requirement	Source
FR-AN-01	KPI tiles: Initial Insurance AR, Total Recovered, Total Outstanding, Recovery Rate (recovered ÷ initial insurance AR).	M
FR-AN-02	Recovered-vs-Outstanding comparison bars by Client, Payer, Panel Type, Denial Category, Agent, and AR Queue (top level).	M
FR-AN-03	Follow-Up Comments Breakdown with tabs Claim Status / Fix-Resolution / Denial Root Cause: # follow-up notes, # distinct claims, current outstanding balance; exportable.	M
FR-AN-04	(N) Add breakdowns by escalation reason category and reassignment reason category once captured.	N
FR-AN-05	All analytics respect scoping (client/clinic/provider viewers see only their data).	BR

32. Reports (FR-RPT)

ID	Requirement	Source
FR-RPT-01	Summary reports (exportable): Denial Analysis Summary (count, outstanding by category), Panel Type Summary (count, outstanding, recovered), AR Aging Summary (six buckets: 0–30, 31–60, 61–90, 91–120, 121–180, 181+), Recovery Performance Summary (recovered vs outstanding by client), Agent Productivity Summary (assigned, completed, recovery \$).	M
FR-RPT-02	AR Collections Progress Summary (mirrors the client report built by hand today): rows = AR queue and sub-queue with totals and grand total; columns Count, Revenue Expectation at 100% / 50% / 30% confidence (initial insurance AR × tier), Yet to Collect at 100/50/30% (open claims only), Payments Posted. Narrative insights: claims in scope and DOS range; worked vs pending; Closed queue realization; Non-Responded value; Denied value incl. Non-Collectible share; overall realization rate. Confidence tiers are master data.	M
FR-RPT-03	(S) The complex ongoing client report (Banuka's Dakota-client report) needs a walkthrough before its logic is specified — open item.	S
FR-RPT-04	Account Manager / Client can generate and export reports within scope (project brief).	N

33. Client Management (FR-CM)

ID	Requirement	Source
FR-CM-01	Client cards: name, eligible claim count, outstanding AR, recovered total, Active/Inactive badge, Activate/Deactivate action (admin).	M
FR-CM-02	Deactivate = stop sync and hide from selectors; data is retained (plan D-15). Reactivation triggers a full re-sync.	M→N
FR-CM-03	(N) Per-client configuration: SharePoint source folder for weekly files, onboarding status, primary account manager.	N

34. User Management (FR-USR)

ID	Requirement	Source
FR-USR-01	User list: name, role, access (human-readable scope), status.	M
FR-USR-02	Add/Edit user: name, role, access level (All Clients / Specific Client / Specific Clinic / Specific Referring Provider); cascading pickers populated from distinct values in the claims data (never free text); validation for each level.	BR
FR-USR-03	(N) Production: SSO/identity integration, deactivate user, manager/team hierarchy (plan T054), audit of permission changes.	N

35. Audit Logs (FR-AUD)

ID	Requirement	Source
FR-AUD-01	Flattened, immutable activity trail across all scoped claims: Timestamp, Claim ID (link), Client, User, Role, Action, Previous Value, New Value, Description. Filters: Search, User, Action, Client. Export.	M
FR-AUD-02	Every state-changing action writes an entry (assignment, reassignment, ad-hoc assignment, follow-up, QA decision, escalation, request resolution, CIP events, auto-adjustment, write-off posting, sync-driven denial changes (N)). System actions use actor “System” with role “Automation” or “ETL”.	M→N

36. System Settings — Master File Maintenance (FR-SET)

Master file	Shape	Notes
Denial Categories	List	Seed: 11 categories (Additional Documentation Required ... Other).
Timely Filing Limit thresholds	Financial class → days	Seed: Commercial 180, Medicare 365, Medicaid 365, Self Pay 9999. Drives tflDeadline and tflRisk.
Non-Collectible Denial Codes	List	Drives Non-Collectible sub-queues. (S) store without CARC prefix.
Panel Types	List	Seed: Toxicology, UTI Panel, STI Panel.
Denial Root Cause Options	List	15 seed values; used when Claim Status = Denied.
Fix / Resolution master	List	14 seed values.
Fix / Resolution by Claim Status	Status → subset of Fix list	Controls options in the follow-up form.
Denial Code → Denial Category mapping	Two columns	Seeded once from the most common category per code in the data; then admin-maintained.
CIP Categories & Required Information	Lists	Seed per Section 22.
Auto-Adjust Denial Codes (N)	List	Session: e.g. 242, 197.
Denial hierarchy / priority ranking (N)	Code → rank	To derive primary denial when mapping is delivered.

Master file	Shape	Notes
Escalation & Reassignment reason categories (N)	Lists	Session request.
Document categories (N)	List	Appeal, EOB, Medical Record, Payer Correspondence, CIP Response, Other.
Denial Category → Recommended Action (N)	Two columns	Currently hard-coded in the mockup.
Revenue confidence tiers (N)	List of %	Seed 100 / 50 / 30.
Action → follow-up TAT (N, Phase 2)	Action → days	Deferred per session.

ID	Requirement	Source
FR-SET-01	List masters are edited as add/remove chip lists with Download Template (CSV) and Upload Template (CSV) for bulk replace; mapping masters are editable tables with the same CSV round-trip.	BR
FR-SET-02	Changes take effect immediately everywhere (filters, forms, classification) without reload; every change is audited (N).	M→N
FR-SET-03	Removing a category does not rewrite existing claims or mappings.	M

37. Shared Table & UI Utilities (FR-UI)

ID	Requirement	Source
FR-UI-01	One shared grid component everywhere: server-side sort, paging (10 / 25 / 50 / 100 rows), record count and selected count, column visibility menu, export of the current filtered and sorted rows, row click to open, optional row selection with an eligibility predicate (ineligible rows render no checkbox and are excluded from Select All).	BR
FR-UI-02	Standard multi-select filter popover with an “All” option on every filter bar.	BR
FR-UI-03	Confirmation dialog before every bulk or irreversible action showing counts and \$ totals; one summary toast after bulk actions.	BR
FR-UI-04	Saved views per user per queue, persisted on the server (mockup uses browser storage).	BR→N
FR-UI-05	Exports are real file downloads (CSV/XLSX) in production — the mockup’s copy-to-clipboard modal is a demo limitation.	N
FR-UI-06	Money formatted \$#,##0.00; dates “MMM DD, YYYY”; light and dark themes supported by the design tokens.	M

38. Excel Export & Bulk Status Upload (FR-XL)

ID	Requirement	Source
FR-XL-01	AR Analyst / AR Manager can export denial line items (claim-level with optional line detail) to Excel from any queue view (project brief).	N

ID	Requirement	Source
FR-XL-02	AR Analyst / AR Manager can upload the exported workbook with updated follow-up fields to update status in bulk; each row is validated against the Comments Framework rules (required fields, Fix-by-Status), conflicts are detected (claim changed since export), and a results summary is shown. Each updated row behaves exactly like a logged follow-up (routes to QA). (Plan T030.)	N
FR-XL-03	Account Manager / Client can export line items and status within scope (read-only; no upload).	N

39. Attachments & Document Vault (FR-DOC)

ID	Requirement	Source
FR-DOC-01	Attachments on follow-up notes (all actions) and CIP client responses; each file has a document category; per-file size limit 10–15 MB (to confirm); allowed types PDF, images, Office documents.	S
FR-DOC-02	Storage in Azure Blob Storage (private container, encryption at rest, access via short-lived signed URLs); metadata (claim, patient account, DOS, category, uploaded by/date, size) in SQL.	S
FR-DOC-03	Document Vault screen: search by claim ID, patient account/name, DOS/visit, category, uploader, date range; scoped like claims.	S
FR-DOC-04	Download and view actions are audited (HIPAA).	S→N

40. Weekly Re-Sync Rules (FR-SYNC)

ID	Requirement	Source
FR-SYNC-01	Implement the scenarios in Section 5 exactly; each sync-driven change writes an audit entry (actor System / ETL) with previous → new denial code and balance.	S
FR-SYNC-02	Store previous primary denial code and a newDenialSinceWork flag to drive the Re-Follow-Up – New Denials sub-queue; clear the flag when the claim is re-assigned and worked.	S→N
FR-SYNC-03	Sync must be idempotent per lab per week (re-running does not duplicate history) and must never overwrite Workbench workflow fields (assignment, notes, QA, CIP).	N

PART C — MOCKUP VS. SESSION DECISIONS

41. Reconciliation Table — Do Not Copy the Mockup As-Is

#	Topic	Mockup behaviour	Build this (decision)
1	Data synced	Seeded claims only; insights drive the Work Queue	Full claim-level master synced weekly; all claims classified (S)
2	Non-Responded rule	denialCategory = “Unresponsive Payers” and no recovery/adjustment	Whole balance in insurance; no payment, denial or adjustment (S)
3	Non-collectible test	Any code in a comma-separated list	Primary denial code only (S)
4	Code format	Prefixed codes (OA257, PR27)	Strip CO/PR/PI/OA for mapping; keep RARC letters (S)
5	Insight grouping	Every code on a claim; all open claims	Primary code; unassigned open claims only (S)
6	Next follow-up default	+7 days	+45 calendar days (S)
7	Re-Follow-Up queue	Single queue	Two sub-queues: Unresolved Claims, New Denials (S)
8	Auto adjustment result	Balance zeroed; claim lands in Closed – Fully Adjusted	Auto Adjustments queue + PMS export + posted confirmation (S)
9	Write-off	Normal QA → Completed	QA → Approved Write-Offs – Pending Posting → agent posts (S)
10	CIP case creation	Pending Approval immediately, in parallel with QA	Visible for approval only after QA approval (S)
11	Expected payment	Label “Expected Payment”; seeded	“Revenue Expectation” = Medicare-rate allowable; 0 if unavailable (S)
12	Potential Recovery	= remaining insurance AR	Revenue Expectation – (Ins Pmt + Pt Pmt + Pt Bal) (S)
13	Recovery Status values	Not Started, In Progress, Partial Recovery, Fully Recovered, Adjusted, Uncollectible, Pending Payer Response, Closed	In Progress, Partially Recovered, Fully Recovered, No Recovery (S) — confirm mapping for Adjusted/Uncollectible
14	Recovered amount	Seeded field	Insurance payments dated after denial date, from master sync (S)
15	Escalate to Supervisor	Also routes claim to QA	Request to TL + Manager; claim routing to confirm
16	Reason categories	Free text only	Category dropdown + text (S)
17	Attachments	CIP client responses only; 3 × 250 KB	All follow-ups + CIP; 10–15 MB; categorized; Document Vault (S)
18	Self-approval	Blocked for agent role only	Blocked for every user (N)
19	Lead approval rights	approve = false in permission table	Lead may approve CIPs (S)

#	Topic	Mockup behaviour	Build this (decision)
20	Persistence	Browser localStorage; session-only batches/users	Server database; all actions audited (N)
21	Exports	Copy-to-clipboard CSV modal	Real CSV/XLSX downloads (N)
22	Line-level denials	Line status/reason only	Expandable per-CPT denial detail (S)

PART D — DATA MODEL

42. Entities & Key Fields

42.1 Claim (claim level)

Group	Fields
Identification	claimId, client (lab), patientAcct, dos, dosEnd, billingProvider, renderingProvider (referring), orderingPractice (clinic), payer, secondaryPayer, financialClass, panel
Ingested status	claimStatus (Fully Denied / Partially Paid / Partially Denied / No Response ...), submissionDate, lastBillDate, paymentDate
Financial	billedAmount, allowedAmount, insurancePayment, patientPayment, adjustmentAmount, insuranceBalance, patientBalance, totalBalance, paymentPct, initialInsuranceAR, remainingAR
Recovery (S)	revenueExpectation, actualPayment, paymentVariance, underpaymentAmount, appealRequired, potentialRecovery, recoveryStatus, recoveredAmount
Denial	primaryDenialCode (normalized, N), denialCodesAll (raw list), denialCategory, denialReason, denialDate, workflowCategory (template key), previousPrimaryDenialCode (N), newDenialSinceWork (N)
Aging & TFL	agingDays, agingBucket, tflDeadline, tflRisk, priority (High/Medium/Low — rule to confirm)
Workflow	workflowStatus, assignedAgent, assignedAgentName, lastFollowUp, nextFollowUp, fixResolution, denialRootCause, workedStatus, adHocFollowUpAssigned, escalationApproved, arQueueTop, arQueueSub (computed)
Adjustment (N)	autoAdjusted, autoAdjustedDate, autoAdjustReasonCode, writeOffApproved, writeOffApprovedDate, pmsPostedConfirmed, pmsPostedDate

42.2 Child and supporting entities

Entity	Key fields
ClaimLine (CPT)	cpt, description, diagnosis, units, billed, allowed, paid, adjusted, denied, balance, status, denialReason, lineDenialCodes[] (raw, with prefix) (S)
FollowUp	date, loggedBy, loggedByRole, claimType, followUpType, claimStatus, denialRootCause, fixResolution, comment, nextFollowUpDate, cip {category, requiredInfo, comment}, attachments[] (N)
Activity (audit)	timestamp, user, role, action, description, previousValue, newValue
QARecord	status (Awaiting QA / QA Approved / QA Rejected), reviewer, isEscalation, agentCompletionDate, errorType, scoreItems (6 booleans), notes
AgentRequest	id, type (Escalation to Supervisor / Reassignment Request), reasonCategory (N), note, requestedBy/Role/Date, status, resolvedBy/Role/Date, resolutionNote
CipCase	id, category, requiredInfo, comment, followUpDate, requestedBy/Role, status, round, clientResponse {text, attachments, respondedBy/Role/Date}, lastReviewDecision {decision, by, byRole, date, note}, history[]

Entity	Key fields
AssignmentBatch	batchId, criteria (client, category, payer, panel, priority, action N), agentCode, createdBy, assignDate, dueDate (optional), claimIds[]
Document (N)	id, claimId, blobUri, fileName, size, contentType, category, uploadedBy, uploadedAt, source (follow-up / CIP)
User & AccessGrant	username, role, agentCode, access {level: all/client/clinic/provider, client, clinic, provider}, status
SavedView	user, queueKey, name, filters JSON
DenialInsight	weekStamp, code, description, count, balance, topPayer, topPayerBalance, impactPct, observation, category, action, claimIds[]
QueueSnapshot (N)	snapshotDate, claimId, arQueueTop, arQueueSub, workflowStatus, balances (nightly, plan T038)
SyncRun (N)	lab, week, startedAt, finishedAt, rowsRead, rowsChanged, errors

PART E — NON-FUNCTIONAL REQUIREMENTS

43. Non-Functional Requirements (NFR)

ID	Requirement	Source
NFR-01	Security: role and scope enforced server-side on every API, export and report; UI hiding is not sufficient.	N
NFR-02	HIPAA: PHI encrypted in transit and at rest; Blob storage private with signed URLs; access and download audit; minimum-necessary PHI in exports; session timeout.	S→N
NFR-03	Auditability: audit entries are append-only and cannot be edited or deleted by any role.	M→N
NFR-04	Multi-tenancy: all data partitioned by lab (client); a user without a grant for a lab can never read its rows.	N
NFR-05	Performance: all grids use server-side filtering, sorting and paging; badge counts and dashboard aggregates computed with indexed queries or cached per sync.	N
NFR-06	Data refresh: weekly sync is idempotent, logged, and recoverable on failure without partial state.	N
NFR-07	Consistency: one shared implementation of lifecycle predicates and classification used by every screen, badge, report and export.	M
NFR-08	Exports: CSV/XLSX with formula-injection protection (escape leading =, +, -, @).	N
NFR-09	Testing: automated end-to-end tests per module plus regression on touched modules; module-by-module handover for testing (S).	BR→S

PART F — PLAN TRACEABILITY & OPEN ITEMS

44. Requirement → Build Plan Traceability

Plan reference: ARWorkflow_Redesign_Plan_v1.0_09.25.2026.xlsx (Plan sheet). Start and end dates have now been filled for every task row from its module's starting row.

Requirement area	Plan module(s) / task(s)	Plan dates
FR-DP Data Processing & Insights	M02 T002–T011	30 Sep – 06 Oct
FR-SET Master files	M03 T012–T020	07 Oct – 12 Oct
FR-WQ Work Queue	M04 T021–T022	12 Oct – 13 Oct
Financial & recovery fields (Sec. 7)	M05 T023–T024	14 Oct – 15 Oct
FR-CD Claim Detail	M06 T025–T027	16 Oct – 18 Oct
FR-FU Follow-up note, FR-XL upload, assignment/export	M06 T028–T032	19 Oct – 21 Oct
FR-ARQ Classification, FR-ADJ auto-adjust	M07 T033–T036	21 Oct – 23 Oct
Events, snapshots, TFL (FR-AUD, FR-SYNC)	M07 T037–T039, M08 T040	23 Oct – 25 Oct
FR-UI Shared table / saved views	M09 T041–T042	25 Oct
FR-MW My Work	M10 T043	25 Oct – 26 Oct
FR-FUM Follow-Up Management	M11 T044–T048	27 Oct – 28 Oct
FR-ACC Roles & permissions	M11 T049–T050	29 Oct – 30 Oct
FR-ACC scoping / FR-USR	M12 T051–T052, M13 T053–T054	02 Nov
FR-ASG Assignment	M14 T055–T058	02 Nov – 04 Nov
FR-QA QA Workflow	M15 T059–T060	04 Nov – 06 Nov
FR-CIP CIP internal	M16 T061–T063	09 Nov – 10 Nov
FR-CLI Client portal	M17 T064–T066	10 Nov – 11 Nov
FR-AUD Audit Logs	M18 T067	11 Nov – 13 Nov
FR-CM Client Management	M19 T068	13 Nov
FR-DASH Dashboard	M20 T069–T070	13 Nov – 15 Nov
FR-AN Analytics	M21 T071–T072	16 Nov – 17 Nov
FR-RPT Reports	M22 T073–T075	17 Nov – 20 Nov
Testing, UAT, go-live	M23 T076–T082	23 Nov – 27 Nov

45. Requirements Not Yet Covered by a Plan Task

These items come from the session or the review and have no task (or no estimate) in the v1.0 plan. Estimates are to be added by the development lead.

#	Requirement	Suggested module	Ref
1	Escalation & Reassignment Requests queue (screen, respond, bulk resolve) — exists in mockup, no dedicated task	M15 / new	FR-REQ
2	Reason categories for escalation and reassignment	M03 + M15	FR-REQ
3	Auto Adjustments queue, PMS export and Mark-as-Posted	M07	FR-ADJ
4	Write-off approval loop and Approved Write-Offs – Pending Posting list	M15	FR-ADJ
5	Re-appearance detection and notification for unposted adjustments/write-offs	M02 / M07	FR-SYNC
6	Re-Follow-Up – New Denials sub-queue and re-sync rules	M07	FR-SYNC
7	Attachments on all follow-ups, Azure Blob storage, Document Vault	New module	FR-DOC
8	CARC prefix normalization and primary-denial derivation (hierarchy table)	M02 / M03	Sec. 4
9	Line-level denial expand in CPT detail	M06	FR-CD
10	Control-plan widget (Open / In Progress / Closed by classification and action) and SLA / Appeal Success metrics	M20	FR-DASH
11	Basic forecasting model (current manual model)	New / M21	Sec. 12
12	Complex client report after walkthrough	M22	FR-RPT
13	Notifications (write-off approved, unposted adjustments, CIP responses)	New	FR-ADJ

Schedule observations: The plan's Go-Live row now ends **27 Nov 2026**, two weeks after the **13 Nov 2026** completion date agreed in the session. Modules M06, M07 and M11 each contain two sub-modules with their own date ranges; undated rows took the dates of the nearest starting row above them within the same module. A later plan version (ARWorkflow_Redesign_Plan_v1.1_09.28.2026.xlsx) also exists in the folder — confirm which version is the baseline.

46. Additional Open Items from the Review

#	Item	Owner
12	Priority (High / Medium / Low) derivation rule — seeded in the mockup, no logic defined	Banuka
13	Aging basis: date of service (mockup) vs. denial date	Banuka
14	Escalate to Supervisor: should the claim also enter QA (mockup) or remain with the agent	Banuka
15	Mapping of mockup recovery statuses (Adjusted, Uncollectible, Pending Payer Response) to the four session statuses	Banuka
16	Auto-Adjust list vs. Non-Collectible list — one master or two	Banuka
17	Keep category workflow-template stage paths on Claim Detail, or show only the four-stage stepper	Banuka / Prasanna

#	Item	Owner
18	TFL thresholds by financial class (180 / 365 / 365) — confirm values per client	Azwin
19	Sub-queue structure for Auto Adjustments (System Auto-Adjusted vs Approved Write-Offs)	Banuka
20	Baseline plan version (v1.0 vs v1.1) and reconciliation of the 27 Nov end date with the 13 Nov commitment	RSKP / Prasanna